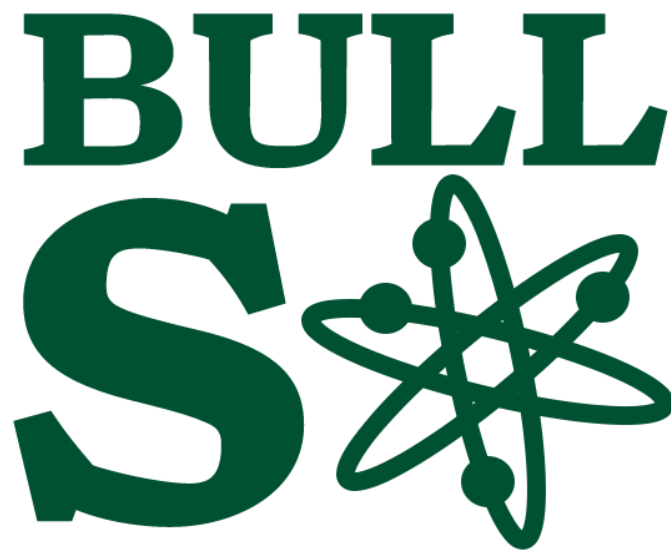


BullSO 2025 Invitational
Optics B



Bull Science Olympiad at the University of South Florida

Saturday, January 25th, 2025

Team Number: _____

Team Name: _____

Participant Names: _____

Total Score: _____/48

Rank: _____

Section 1. Multiple Choice. (1 point each)

1. What is the primary effect of a convex lens?
 - a. To diverge light rays
 - b. To converge light rays
 - c. To filter out specific wavelengths
 - d. To create a virtual image

2. Which type of curved mirror always creates a virtual and upright image?
 - a. Plane mirror
 - b. Concave mirror
 - c. Convex mirror
 - d. None of the above

3. Which phenomenon is responsible for the dispersion of light in a prism?
 - a. Reflection
 - b. Refraction
 - c. Diffraction
 - d. Polarization

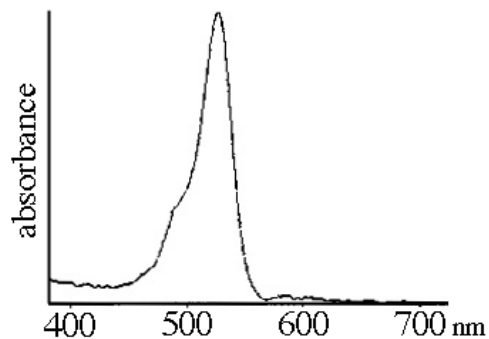
4. What happens to light when it passes from a medium of higher refractive index to a lower refractive index at an angle greater than the critical angle?
 - a. The light is absorbed
 - b. The light refracts into the second medium
 - c. The light undergoes total internal reflection
 - d. The light passes through without bending

5. Which of the following describes the focal point of a concave mirror?
 - a. The point where parallel light rays converge after reflection
 - b. The point where light rays appear to originate after reflection
 - c. The center of curvature of the mirror
 - d. The distance between the object and the mirror

6. What is the unit of measurement for the power of a lens?
 - a. Diopters
 - b. Meters
 - c. Watts
 - d. Joules
7. Which phenomenon explains the creation of a rainbow?
 - a. Diffraction of light
 - b. Polarization of light
 - c. Dispersion of light through water droplets
 - d. Absorption of light by air molecules
8. The “red-eye” effect in photographs can be attributed to light bouncing off which part of the eye?
 - a. Choroid
 - b. Retina
 - c. Sclera
 - d. Optic Disc
9. What type of image is formed by a convex lens when the object is placed beyond the focal point?
 - a. Virtual and magnified
 - b. Real and inverted
 - c. Virtual and inverted
 - d. Real and upright
10. Which optical device is used in a periscope to change the direction of light?
 - a. Lenses
 - b. Plane mirrors
 - c. Convex mirrors
 - d. Concave lenses
11. Which of the following uses an additive color model?
 - a. Computer monitors
 - b. Printers
 - c. Silk-screening
 - d. Painting

12. What lens is likely to be used in a patient with myopia?
- Converging lens
 - Diverging lens
 - Spherical lens
 - Parabolic lens
13. Which of the following controls the shape of the lens in the human eye?
- Retina
 - Cornea
 - Choroid
 - Ciliary Body
14. Which of the following is a subtractive color?
- Red
 - Blue
 - Green
 - Yellow
15. Which of the following is the most common cause of visual impairment worldwide?
- Cataracts
 - Glaucoma
 - Age-related Macular Degeneration
 - Congenital blindness
16. Which of the following is NOT true of focal length?
- It measures how strongly light is diverged or converged
 - It is the inverse of optical power
 - A negative focal length indicates that the system converges light
 - Longer focal length leads to higher magnification
17. What happens to the speed of light as it enters a denser medium?
- It increases
 - It decreases
 - It remains unchanged
 - It becomes equal to the speed of sound

18. What is the purpose of anti-reflective coatings on lenses?
- To block harmful UV light
 - To increase the reflectivity of light
 - To reduce the reflection of light and increase transmission
 - To enhance image magnification
19. What wavelengths of light are generally subject to lower indexes of refraction?
- Shorter wavelengths
 - Longer wavelengths
 - All wavelengths have the same index of refraction
 - It is random
20. Cones found in the human eye can be categorized as:
- Small, medium, large
 - Short, medium, long
 - Short or long
21. The following absorption spectrum was obtained in a lab from a sample of food dye. What color dye is likely to be present?



- Blue
- Yellow
- Violet
- Red

Section 2. Short answer. Please show accompanying work if the question requires a numerical solution. (3 pts each)

22. Explain how light travels through optical fibers.

23. What is the critical angle of an interface composed of air and honey ($n = 1.9$)?

24. If a ray of light passes from air to a pane of glass ($n = 1.5$), and the ray enters the glass pane with an incident angle of 30 degrees, what angle will the refracted ray make with the normal?

25. List the four colors used by color printers and state what color model is used for printing.

Section 3. Long answer. Please write in complete sentences. (5 pts each)

26. Define Snell's Law and write its equation.

27. Describe the difference between a real image and a virtual image.

28. Select three of the following structures in the human eye. Describe its function or characteristics in the body.

Lateral Rectus Muscle	Optic Nerve	Lacrimal Caruncle	Choroid
Anterior Chamber	Vitreous Body	Cones	